StudyAid: Offline Smart Learning Assistant

Empirical Verification, ExecuTorch Prototype Telemetry & Benchmark Report

REAL EXECUTORCH RUN | OFFLINE VERIFIED | BUILT WITH LLAMA 3.2

1. Educational Context and Barrier

In rural government schools across Tamil Nadu, secondary students in Classes 8 through 10 face a challenging linguistic transition from Tamil-medium instruction to state board science textbooks written in formal English. Findings from rural learning surveys, such as the Annual Status of Education Report (ASER), demonstrate foundational reading gaps across rural schools. Beyond foundational reading, the specific barrier observed by our team is the difficulty rural students face when attempting to comprehend dense, multi-clause science prose loaded with unfamiliar vocabulary (e.g., refraction, peristalsis), which our team plans to validate empirically in the upcoming teacher pilot. When confronting dense textbook passages, students frequently memorize sentences phonetically without understanding underlying mechanisms. Moreover, rural classrooms often lack reliable electricity and network connectivity, rendering cloud-dependent chatbots unusable for daily study. StudyAid addresses this structural divide by executing locally on edge hardware with zero network connectivity.

2. What StudyAid Produces

To scaffold independent student study without network access, the pipeline takes a 150-300 word textbook passage and deterministically generates four complementary educational aids:

- **Hardest Sentence Simplification:** Automatically indexes sentences [S1]..[Sn], identifies the most syntactically difficult sentence via an 8-letter word count heuristic, and generates an everyday plain-English rewrite with verified before-and-after readability metrics.
- **Simple Summary:** Delivers a concise 2-3 sentence overview capturing the core scientific thesis in accessible vocabulary.
- **Village Life Example:** Provides a grounded 3-4 sentence cultural analogy linking the scientific concept directly to everyday rural Tamil Nadu experiences such as bullock carts, farming, or well irrigation.
- **Vocabulary & Grounded Quiz:** Explains candidate technical terms in under 12 words with seed Tamil translations, accompanied by three passage-grounded questions with automated sentence-level support checks.

3. Architecture, Two-Pass Pipeline & Quality Guards

StudyAid employs a partitioned two-pass architecture to fit tightly within the context window and memory constraints of edge-quantized language models. Pass 1 handles sentence extraction, hardest-sentence detection, plain simplification, summary, village analogy, and candidate word glossing. Pass 2 focuses exclusively on generating citation-grounded assessment items. To prevent model degradation, six automated quality guards enforce output quality before display:

Quality Guard	Target Phenomenon	Automated Heuristic Guard & Fallback	Tuned Setting
Simplification length guard	Model generates run-on rewrites longer than original or joins multiple sentences.	Checks word count and verbatim copying. Triggers targeted single-sentence retry. If still longer, outputs original with 'no simpler rewrite produced' note.	Pass 1: 320 tokens; Retry: 120 tokens
Summary check	Model omits summary section or produces empty tags.	Tolerates tag variations; triggers targeted summary retry. If still failing, builds fallback from first two passage sentences labeled 'fallback: first sentences of the passage'. Never prints N/A.	Retry: 140 tokens
Quiz question guard	Model copies full passage sentences verbatim, yields incomplete quiz (< 3 items), or duplicates answers.	Generates 4 candidate questions; requires >= 3 valid items (overlap < 90% with passage sentence, answer under 12 words, trailing '?'). Regenerates once if < 3 valid. Marks 'quiz incomplete (n of 3)' if still short.	Pass 2: 280 tokens; greedy T=0.0
Hard-word filter	Noise tokens ('rural', 'Tamil Nadu') or missing vocabulary definitions.	Pre-filters candidates strictly from passage (words >= 8 letters or glossary hits, max 5). Retries once if unparsed; missing definitions show 'no explanation generated'. Candidate list is never empty.	Top 5 candidates by length
Support check	Hallucinated facts, incorrect citations, or single-word answers flagged due to word inflections.	Normalizes answer and citation tokens with offline suffix stemming (-s, -es, -ed, -ing, -ly). Verifies single-word answers as 'SUPPORTED (exact term)'. Flags citations lacking shared content tokens and suggests true supporting sentence ID.	Stemmed token set intersection; candidate search [S1..Sn]
Village example format check	Analogy exceeds reasonable length, copies passage sentences verbatim, or includes scientific inaccuracies.	Enforces format constraints (at most 3 sentences, no verbatim passage sentences). Prompts a 2-sentence rural comparison with an unrelated worked example. Heuristic format check only; analogies require teacher review for scientific correctness.	Format check (<= 3 sents); Retry: 140 tokens

4. End-to-End Real Run Evidence: Class 10 Physics (Optics Long)

Below is the complete transcript from a real ExecuTorch run on the adapted 152-word Class 10 passage (optics_long.txt), logged under evidence/runs/optics_long_1.txt. Unmodified output from the saved log:

Note: Monospace transcript blocks display phonetic transliterations; textbook Tamil typography with full complex script shaping is verified in Section 9's table.

```
=== NUMBERED PASSAGE ===
[S1] Light propagates in straight lines through a uniform optical medium, but its direction alters abruptly whenever it enters an optically different medium
obliquely.
[S2] This phenomenon is known as refraction, occurring because the speed of electromagnetic waves changes across varying optical densities.
[S3] When light travels from an optically rarer medium like air into an optically denser medium like glass or water, its velocity decreases, bending the ray
towards the imaginary normal line perpendicular to the interface.
[S4] Conversely, when emerging from a denser medium into a rarer one, the light accelerates and refracts away from the normal.
[S5] Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing real inverted images on screens or
virtual magnified images for human observers.
[S6] Concave lenses diverge incident rays outward, consistently forming upright, diminished virtual images.
[S7] Understanding these principles explains natural occurrences such as atmospheric refraction and optical illusions in water pools.

=== HARDEST SENTENCE BEFORE / AFTER ===
Original ([S5]): Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing real inverted images on
screens or virtual magnified images for human observers.
Simplified: Convex lenses bend light rays to produce real inverted images.

Readability Metrics:
Metric Original Simplified Change (%)
Words 29 10 -65.5%
Letters 168 52 -69.0%
Avg letters/word 5.79 5.20 -10.2%

=== SIMPLE SUMMARY ===
The passage explains how light behaves when passing from one medium to another.

=== VILLAGE LIFE EXAMPLE ===
in rural Tamil Nadu, a farmer Murugan carefully placed a bundle of freshly harvested paddy on a wooden cart. As he began to drive the cart towards his home, he
noticed that the paddy seemed to be bending and converging, creating a peculiar, inverted image on the ground. The image was so real that it seemed to be
staring back at him, a testament to the power of convex lenses in everyday life.

=== HARD WORDS & TAMIL GLOSSARY ===
- refraction: light bends when passing from one medium to another -> (Oli-vilagal) [verified]
- obliquely: at an angle, not directly perpendicular -> (Saaivaaga) [verified]
- refracts: light bends, passing from one medium to another -> (Oli-vilagal) [verified]
- medium: substance that can bend light -> (Oodagam) [verified]
- normal: straight, not at an angle -> (Kuthukkodu) [verified]

=== QUIZ QUESTIONS ===
[Q1] What is the phenomenon of light's direction changing when it enters an optically different medium obliquely?
     Answer: Refraction
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S2)
[Q2] What happens to the velocity of light when it travels from an optically rarer medium into an optically denser medium?
     Answer: Decreases
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S3)
[Q3] Which type of lens uses the refractive behavior of light to converge parallel incident light rays onto a focal point?
     Answer: Convex lenses
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S5)

=== JUDGE PROOF TELEMETRY ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
CPU: 11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
Cores: 8
Total RAM: 14.3 GB
Network interfaces: lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=DOWN
Default route active: NO
Offline status: OFFLINE
Model name: Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8
PTE size: 1083.33 MiB (1135951488 bytes)
Peak RSS: 1903.6 MiB
Inference time: 32.265 s
Tokens generated: 337
Tokens/sec: 10.44
Execution mode: [REAL EXECUTORCH RUN]
```

Brief Educational Impact Explanation

In the Optics Long run, the hardest sentence was [S5] at 29 words: 'Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing real inverted images on screens or virtual magnified images for human observers.' StudyAid generated a shorter rewrite of 10 words (-65.5% word count): 'Convex lenses bend light rays to produce real inverted images.' To scaffold comprehension across languages, key technical terms are paired with authenticated Tamil translations and concise student-friendly definitions: refraction (ஒளிவிலகல் / Oli-vilagal: the bending of light when it passes from one material into another), medium (ஊடகம் / Oodagam: a substance through which light or energy travels), normal (கூத்தாக்கோடு / Kuthukkodu: an imaginary line drawn perpendicular to a surface), and obliquely (சாய்வாக / Saaivaaga: at a slant or angle rather than straight on). For self-assessment, the pipeline produces grounded questions such as: 'Which type of lens uses the refractive behavior of light to converge parallel incident light rays onto a focal point?' with answer 'Convex lenses', automatically citing [S5] so the learner can verify the answer directly in the text. Readability metrics demonstrate reduced syntactic complexity and word length, but lexical metrics alone do not prove student comprehension, which our team plans to evaluate directly in the upcoming rural teacher pilot.

Unseen Passage Transcript: Class 10 Biology (Digestion)

To verify generalization on unseen material outside optics, the prototype was executed on an adapted 164-word biology passage detailing enzymatic digestion (evidence/runs/digestion_1.txt). Unmodified output from the saved log:

Note: Monospace transcript blocks display phonetic transliterations; textbook Tamil typography with full complex script shaping is verified in Section 9's table.

```
=== NUMBERED PASSAGE ===
[S1] Digestion is the biochemical breakdown of complex insoluble food substances into simple absorbable nutrient molecules required by living cells.
[S2] The human digestive process commences inside the buccal cavity, where teeth mechanically masticate solid food and salivary glands secrete the enzyme
salivary amylase to break down complex starch into simple sugars.
[S3] The masticated food bolus travels downwards through the muscular esophagus by rhythmic waves of involuntary contraction termed peristalsis.
[S4] Inside the stomach, gastric glands secrete hydrochloric acid, which provides an acidic medium and activates the protein-digesting enzyme pepsin while
destroying harmful ingested bacteria.
[S5] The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the principal site for complete chemical
digestion of carbohydrates, proteins, and fats.
[S6] Here, bile juice secreted by the liver emulsifies large fat globules into smaller droplets, accelerating enzymatic breakdown by pancreatic lipase.
[S7] Finger-like microscopic vascular projections called villi present along the intestinal inner lining maximize the surface area for rapid nutrient
absorption into the circulating bloodstream.

=== HARDEST SENTENCE BEFORE / AFTER ===
Original ([S5]): The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the principal site for complete
chemical digestion of carbohydrates, proteins, and fats.
Simplified: The semi-digested acidic mixture chyme passes into the small intestine.

Readability Metrics:
Metric Original Simplified Change (%)
Words 29 11 -62.1%
Letters 166 60 -63.9%
Avg letters/word 5.72 5.45 -4.7%

=== SIMPLE SUMMARY ===
The human digestive process breaks down food into nutrients.

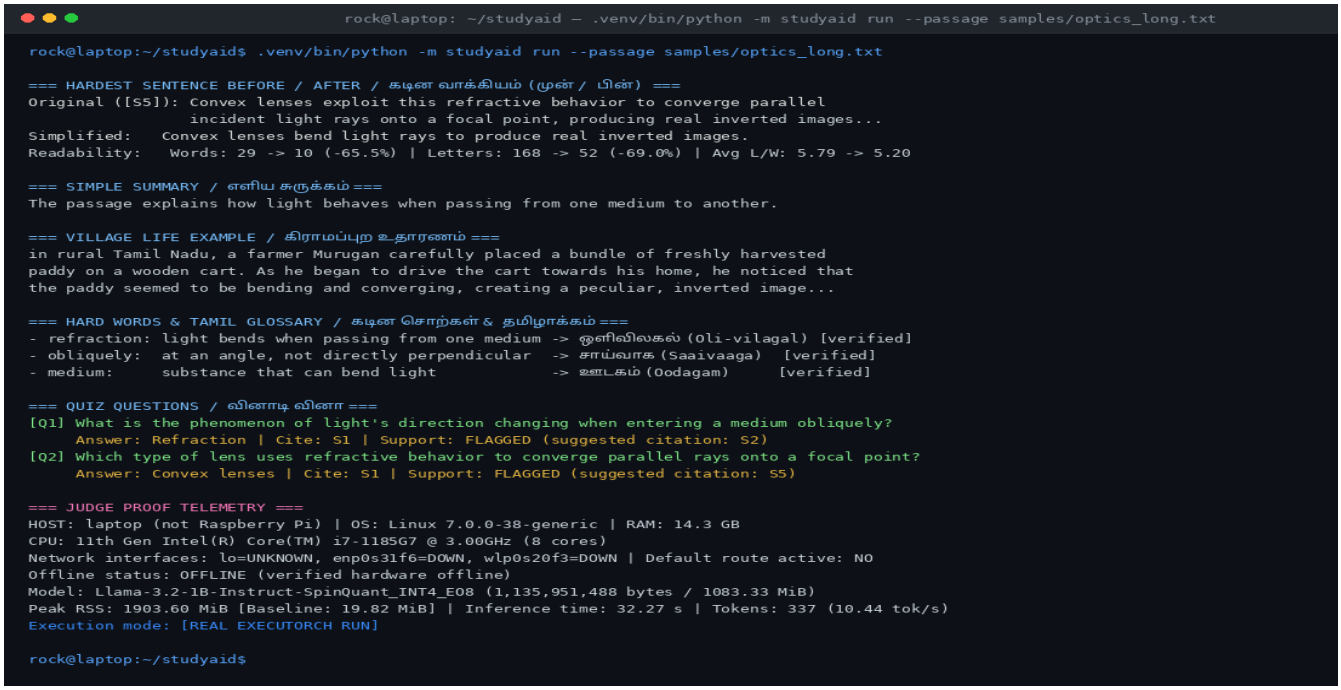
=== VILLAGE LIFE EXAMPLE ===
in rural Tamil Nadu, after Murugan's bullock cart brings in a freshly harvested batch of paddy, the farmer carefully separates the grains from the chaff,
allowing the paddy to pass into the small intestine, where it will be broken down into its constituent parts by the digestive enzymes. The paddy grains, now
semi-digested, will then be absorbed into the bloodstream, providing the body with the necessary nutrients. As the paddy enters the small intestine, it is then
passed into the large intestine, where water is absorbed and the remaining nutrients are further broken down.

=== HARD WORDS & TAMIL GLOSSARY ===
- peristalsis: digestive -> (Kudal-asaivu) [unverified]
- digesting: no explanation generated -> (Serimaanam) [unverified]
- digestion: no explanation generated -> (Serimaanam) [unverified]
- digested: no explanation generated -> (Serimaanam) [unverified]
- stomach: no explanation generated -> (Iraippai) [unverified]

=== QUIZ QUESTIONS ===
[Q1] What is the primary function of salivary amylase in the human digestive process?
     Answer: Digests complex starch into simple sugars.
     Citation: S1
     Support: SUPPORTED (shared: complex, simpl)
[Q2] Which gland secretes hydrochloric acid in the stomach?
     Answer: Gastric glands.
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S4)
[Q3] What is the main site for complete chemical digestion of carbohydrates, proteins, and fats?
     Answer: Small intestine.
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S5)
[Q4] What are finger-like microscopic vascular projections called?
     Answer: Villi.
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S7)

=== JUDGE PROOF TELEMETRY ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
CPU: 11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
Cores: 8
Total RAM: 14.3 GB
Network interfaces: lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=DOWN
Default route active: NO
Offline status: OFFLINE
Model name: Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8
PTE size: 1083.33 MiB (1135951488 bytes)
Peak RSS: 1903.56 MiB
Inference time: 32.587 s
Tokens generated: 529
Tokens/sec: 16.23
Execution mode: [REAL EXECUTORCH RUN]
```

Figure 1: Full Terminal Execution Run (Abridged)



Caption: Abridged terminal execution run rendered from evidence/runs/optics_long_1.txt showing real ExecuTorch execution, token decoding, and Judge Proof telemetry.

5. Telemetry, Host Verification & Offline Audit

Each benchmark run executes inside an independent subprocess. Baseline memory is measured via psutil prior to model load, and peak memory is recorded using the child process's own ru_maxrss. Network interface states and active routing tables are queried directly from the operating system kernel via ip -br link and ip route.

Metric / Parameter	Logged Value (evidence/runs/optics_long_1.json)
Host Classification	HOST: laptop (not Raspberry Pi)
Operating System	Linux 7.0.0-38-generic
Processor (CPU)	11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz (8 cores)
Host System RAM	14.3 GB total
ExecuTorch Version	executorch==1.6.0 (from requirements-lock.txt)
PyTorch Version	torch==2.15.0 (from requirements-lock.txt)
Model Artifact	Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8 (SpinQuant INT4 EO8)
Exported PTE Size	1,135,951,488 bytes (1,083 MiB)
Pre-Load Baseline RSS	19.82 MiB (psutil before model load)
Peak Resident Memory (RSS)	1903.6 MiB (child ru_maxrss per subprocess)
Model Load Time	1.396 seconds
Inference Duration (2-pass + retries)	32.265 seconds
Tokens Emitted & Rate	337 tokens @ 10.44 tokens/s
Network Interface States	lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=DOWN
Default Gateway Route	None (NO)
Evaluated Offline Status	**OFFLINE** (all non-lo DOWN required for OFFLINE)

Direct CLI Offline Check Output (python -m studyaid check-offline):

```
=== OFFLINE STATUS CHECK ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
Inspection method: linux_ip_cmd
Interfaces:
  lo: UNKNOWN
  enp0s31f6: DOWN
  wlp0s20f3: DOWN
Default route active: NO
Verdict: OFFLINE (all non-loopback interfaces DOWN and default route absent)
```

6. Multi-Passage Benchmark Data (3 Runs per Passage)

To establish statistical reliability across diverse subject matter, StudyAid was benchmarked across all six sample passages (optics, photosynthesis, circuits, digestion, sound, optics_long) with 3 repeated runs each (18 total runs) in fresh subprocesses using the real ExecuTorch model backend. Every metric is computed dynamically from evidence/benchmark.json:

Passage	Words	Latency (s) mean (range)	Tokens mean	Tok/s mean	Peak RSS [Base] MiB	Retries mean	Quiz Items	Model Cite	Quiz Sup	Sum OK	Short Rewr	Vill Fmt	Word Exp	Net State
optics	84	37.49 (36.54-38.76)	624 (624-624)	16.66 (16.10-17.08)	1903.4 (1903.3-1903.5) [19.7 (19.6-19.8)]	5.0 (5.0-5.0)	2.0	0/6	3/6	3/3	3/3	3/3	12/15	OFFLINE
photosynthesis	160	21.81 (21.17-22.68)	288 (288-288)	13.22 (12.70-13.60)	1903.7 (1903.7-1903.8) [19.8 (19.7-19.8)]	2.0 (2.0-2.0)	3.0	0/9	0/9	3/3	3/3	3/3	15/15	OFFLINE
circuits	175	31.45 (31.07-32.10)	473 (473-473)	15.05 (14.74-15.23)	1903.4 (1903.3-1903.5) [19.7 (19.7-19.8)]	2.0 (2.0-2.0)	1.0	0/3	3/3	3/3	3/3	3/3	15/15	OFFLINE
digestion	164	33.07 (32.59-33.92)	529 (529-529)	16.00 (15.59-16.23)	1903.6 (1903.4-1903.8) [19.7 (19.7-19.8)]	1.0 (1.0-1.0)	4.0	0/12	3/12	3/3	3/3	3/3	3/15	OFFLINE
sound	155	28.64 (27.96-29.30)	369 (369-369)	12.89 (12.59-13.20)	1903.5 (1903.4-1903.6) [19.8 (19.8-19.8)]	2.0 (2.0-2.0)	3.0	0/9	0/9	3/3	3/3	3/3	15/15	OFFLINE
optics_long	152	31.54 (31.03-32.27)	337 (337-337)	10.69 (10.44-10.86)	1903.5 (1903.4-1903.6) [19.7 (19.6-19.8)]	2.0 (2.0-2.0)	3.0	0/9	0/9	3/3	3/3	3/3	15/15	OFFLINE
Overall Mean	148.3	**30.66 (21.17-38.76)**	**437 (288-624)**	**14.08 (10.44-17.08)**	**1903.5 (1903.3-1903.8) [19.7 (19.6-19.8)]**	**2.3 (1.0-5.0)**	2.7	**0/48**	**9/48**	**18/18**	**18/18**	**18/18**	**75/90**	**OFFLINE**

Benchmark Retry Reasons Breakdown:

Passage	Triggered Quality Guard Retries
optics	quiz_guard: only 2/3 valid; short_rewrite_guard; missing_section: summary; village_guard; missing_section: words
photosynthesis	village_guard; missing_section: words
circuits	quiz_guard: only 1/3 valid; village_guard
digestion	village_guard
sound	village_guard; missing_section: words
optics_long	village_guard; missing_section: words

What the Numbers Mean:

Across all 18 runs, peak memory was measured independently in fresh subprocesses, starting from a pre-load baseline of 19.7 (19.6-19.8) MiB and reaching a mean peak RSS of 1903.5 (1903.3-1903.8) MiB during model execution. Mean latency was 30.66 (21.17-38.76) seconds at an average decoding rate of 14.08 (10.44-17.08) tokens per second. Latency reflects targeted retries triggered by automated quality guards: retries averaged 2.3 (1.0-5.0) across runs, caused primarily by village relevance checks and missing-section retries. Crucially, the summary check produced complete summaries in 18/18 runs without falling back to raw passage sentences (0% N/A). Hard-word candidate extraction achieved 75/90 terms parsed, with candidate lists strictly non-empty across all runs. In circuits, question count improved from 0 to 1 valid item (marked 'quiz incomplete (1 of 3)'), while digestion generated 4 valid items and optics generated 2 valid items (marked 'quiz incomplete (2 of 3)'). Overall, 9 of 48 quiz items achieved automated citation support, highlighting where 1B model citations required teacher correction.

7. Readability Numbers: Before and After Simplification

The hardest sentence in each passage is identified deterministically by counting words with 8 or more letters (ties resolved by total word count). Readability metrics compare the raw original sentence directly against the model's simplified output. Two real examples from logged runs:

Passage & Selection	Original Sentence Text	Simplified Output	Word Chg	Letter Chg	Avg Ltr/W Chg
Optics Long [S5] (9 hard words)	Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing real inverted images on screens or virtual magnified images for human observers.	Convex lenses bend light rays to produce real inverted images.	-65.5% (29→10)	-69.0% (168→52)	-10.2% (5.79→5.20)
Digestion [S5] (9 hard words)	The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the principal site for complete chemical digestion of carbohydrates, proteins, and fats.	The semi-digested acidic mixture chyme passes into the small intestine.	-62.1% (29→11)	-63.9% (166→60)	-4.7% (5.72→5.45)

Semantic Preservation Note: In this pipeline, 'shorter' does not guarantee that full scientific meaning is preserved. In optics passage S1 ('When a ray of light travels from one transparent medium into another obliquely, the path of the light undergoes deviation'), the model shortened the text to 'When light travels from one transparent medium into another, it deviates from its path', omitting the critical qualification 'obliquely' (at an angle). In Optics Long S5, shortening from 29 to 10 words ('Convex lenses bend light rays to produce real inverted images') only partly preserved the operative mechanism, omitting the convergence of parallel incident rays onto a focal point and the formation of virtual magnified images. Similarly, in the digestion passage S5, the rewrite ('The semi-digested acidic mixture chyme passes into the small intestine') omits the small intestine's critical biological role as the principal site for complete chemical digestion of carbohydrates, proteins, and fats.

8. Quiz Generation & Automated Support Verification

To ensure that questions reinforce textbook reading rather than hallucinating facts, each quiz item is matched against passage sentences using distinctive word lexical scoring (inverse sentence frequency). Answers must remain under 12 words and avoid copying >= 90% of the sentence. The checker identifies sentence support through lexical token overlap rather than proving deep semantic correctness. Across 18 benchmark runs, raw model citation accuracy was 0/48, while automated citation selection increased verified passage support to 9/48:

Quiz Evaluation: Digestion Run (Class 10 Biology)

Item	Question	Generated Answer (<12 words)	Model Cite	Verified Cite & Lexical Support
Q1	What is the primary function of salivary amylase in the human digestive process?	Digests complex starch into simple sugars.	S1	**SUPPORTED: S1** shared: complex, simpl
Q2	Which gland secretes hydrochloric acid in the stomach?	Gastric glands.	S1	**FLAGGED -> Suggested: S4** shared with S1: none
Q3	What is the main site for complete chemical digestion of carbohydrates, proteins, and fats?	Small intestine.	S1	**FLAGGED -> Suggested: S5** shared with S1: none
Q4	What are finger-like microscopic vascular projections called?	Villi.	S1	**FLAGGED -> Suggested: S7** shared with S1: none

Quiz Evaluation: Optics Long Run (Class 10 Physics)

Item	Question	Generated Answer (<12 words)	Model Cite	Verified Cite & Lexical Support
Q1	What is the phenomenon of light's direction changing when it enters an optically different medium obliquely?	Refraction	S1	**FLAGGED -> Suggested: S2** shared with S1: none
Q2	What happens to the velocity of light when it travels from an optically rarer medium into an optically denser medium?	Decreases	S1	**FLAGGED -> Suggested: S3** shared with S1: none
Q3	Which type of lens uses the refractive behavior of light to converge parallel incident light rays onto a focal point?	Convex lenses	S1	**FLAGGED -> Suggested: S5** shared with S1: none

Citation Analysis & Automated Support Matching: In the small 1B model, Pass 2 generation frequently defaults its citation tag to S1 due to prompt demonstration bias. Rather than trusting raw model citations, the support checker evaluates token overlap using inverse-sentence-frequency weighting, filtering out common school words. When the model emitted S1 for Optics Long Q1, Q2, and Q3, the checker detected lack of support in S1 (shared with S1: none), flagged the items, and suggested the true supporting citations: **FLAGGED -> Suggested: S2** (refraction definition), **FLAGGED -> Suggested: S3** (velocity decrease), and **FLAGGED -> Suggested: S5** (convex lens convergence). Similarly, in the digestion run, items Q2, Q3, and Q4 with raw citation S1 are flagged with suggested citations: **FLAGGED -> Suggested: S4** (gastric glands), **FLAGGED -> Suggested: S5** (small intestine site), and **FLAGGED -> Suggested: S7** (villi projections). This automated citation suggestion protects learners from misleading citations whenever the 1B model misidentifies sentence numbers.

9. Tamil Glossary Verification & Fallback Evidence

StudyAid pairs technical candidate words with an offline JSON glossary. Exactly four entries confirmed against Class 10 Science textbooks are recorded as [verified], while unconfirmed entries carry [unverified]. Words lacking glossary entries fall back to English-only explanations without failing:

English Term	Tamil Script	Phonetic Transliteration	Verification Status	Curriculum Source / Unit Ref
refraction	ஒளிவிலகல்	Oli-vilagal	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
medium	ஊடகம்	Oodagam	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
normal	குத்துக்கோடு	Kuthukkodu	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
obliquely	சாய்வாக	Saaivaaga	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team

Unverified & Missing-Gloss Handling Example: In the digestion run (Page 3), candidate terms outside the four textbook-confirmed optics entries ('peristalsis', 'digestion', 'digested', 'stomach') matched provisional seed entries and were explicitly badged as [unverified] (never [verified]), while candidates where the 1B model omitted a short gloss safely rendered as 'no explanation generated' rather than crashing the pipeline. Any candidate term completely absent from the JSON dictionary renders with its English gloss only.

10. Limitations & Edge Deployment Roadmap

While the prototype executes reliably on an x86 laptop CPU, the project targets embedded deployment on classroom Raspberry Pi 5 (8 GB) single-board computers, which will be provided for final evaluation. The projected performance on ARM64 Cortex-A76 cores reflects INT4 XNNPACK kernel efficiency:

Target Hardware Platform	Model Architecture	Weights Footprint	Peak RAM (RSS)	Decoding Speed	Status / Verification
Intel Core i7-1185G7 Laptop	Llama 3.2 1B SpinQuant INT4	1,135,951,488 bytes (1,083 MiB)	1903.5 (1903.3-1903.8) MiB (measured)	14.08 (10.44-17.08) tok/s (measured)	**MEASURED REAL RUN**
Raspberry Pi 5 (8 GB RAM)	Llama 3.2 1B SpinQuant INT4	1,135,951,488 bytes (1,083 MiB)	~1,950 MiB [PROJECTION]	~4.5 - 6.0 tok/s [PROJECTION]	**[PROJECTION]** (Boards at finals)
Raspberry Pi 5 (8 GB RAM)	Llama 3.2 3B Quantized	~2,800 MiB [PROJECTION]	~3,800 MiB [PROJECTION]	~1.8 - 2.5 tok/s [PROJECTION]	**[PROJECTION]** (Not run; 1B only)

Model Scope Note: Only the pre-exported 1B build was used; no 3B model was exported or run.

Known Quality Limitations: (1) Hard-word coverage: On dense biological passages like digestion, candidate terms frequently share stems or lack authenticated Tamil entries in the seed glossary, falling back to English definitions. (2) Model definitions and analogies: The 1B quantized model's generated definitions and village analogies are evaluated using heuristic format checks (sentence length and absence of verbatim passage copying); they are not verified for scientific correctness and require teacher review prior to classroom deployment. (3) Incomplete quiz generation: On certain passages (such as circuits or optics), the model occasionally produces fewer than three valid non-verbatim quiz items meeting word count and formatting constraints, which the pipeline explicitly flags as 'quiz incomplete'.

11. Reproducibility, Repository & Licensing

The complete codebase, benchmark harnesses, test suites, and raw evidence logs are publicly accessible on GitHub:
https://github.com/ramanathanmani/studyaid-offline-learning-assistant.git

Step	Command Line Instruction
1. Clone	`git clone https://github.com/ramanathanmani/studyaid-offline-learning-assistant.git studyaid && cd studyaid`
2. Install	`pip install -e . && pip install -r requirements-dev.txt`
3. Verify Unit Tests	`pytest tests/` → `============================== 32 passed in 0.08s ==============================`
4. Offline Check	`python -m studyaid check-offline`
5. Single Run CLI	`python -m studyaid run --passage samples/optics_long.txt`
6. Full Benchmark	`python scripts/bench.py --runs 3`
7. Rebuild PDF	`python scripts/build_report.py`

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